INCH-POUND

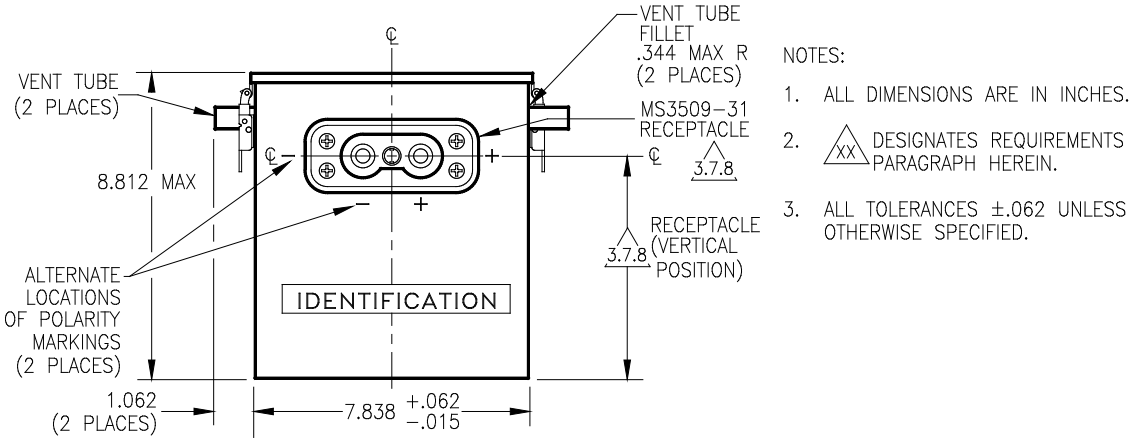
MIL-PRF-81757/11E
w/AMENDMENT 1
5 August 2021
SUPERSEDING
MIL-PRF-81757/11E(AS)
1 March 1999

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, 24-VOLT, 20-AMPERE-HOUR, LOW-FREQUENCY-VIBRATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.



FRONT VIEW

FIGURE 1. Dimensions and configuration for PIN M81757/11-3.

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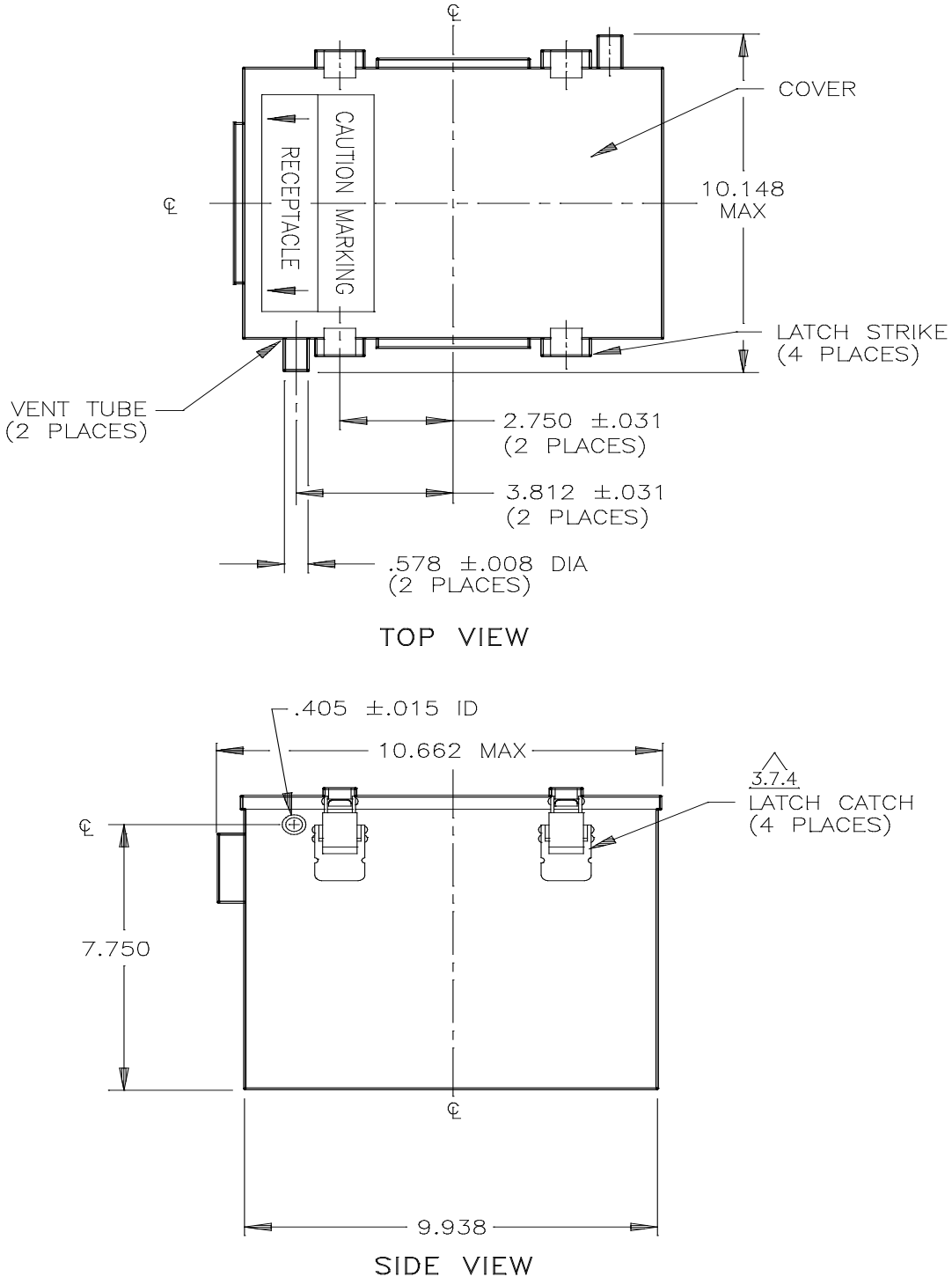
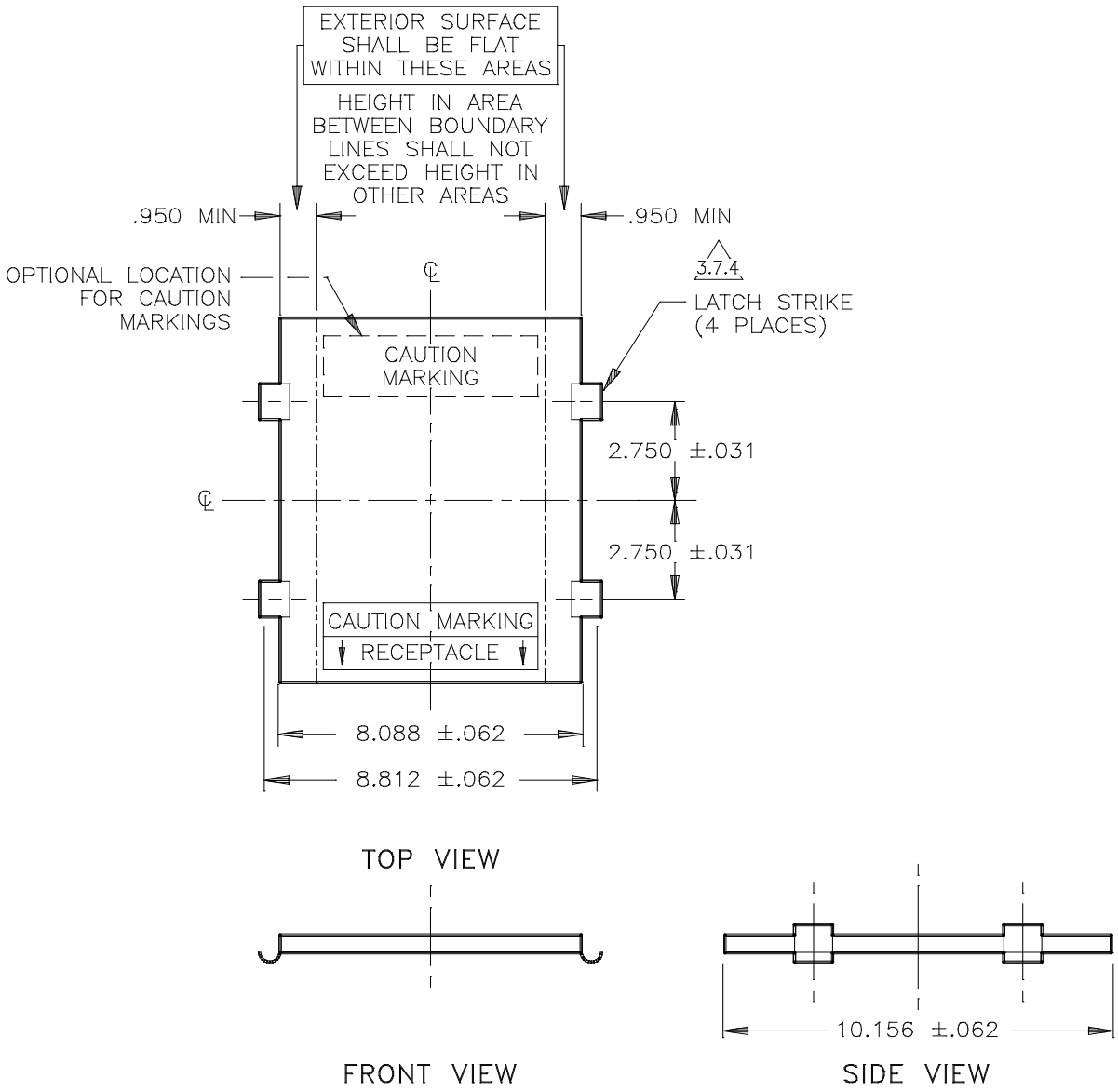


FIGURE 1. Dimensions and configuration for PIN M81757/11-3 - Continued.

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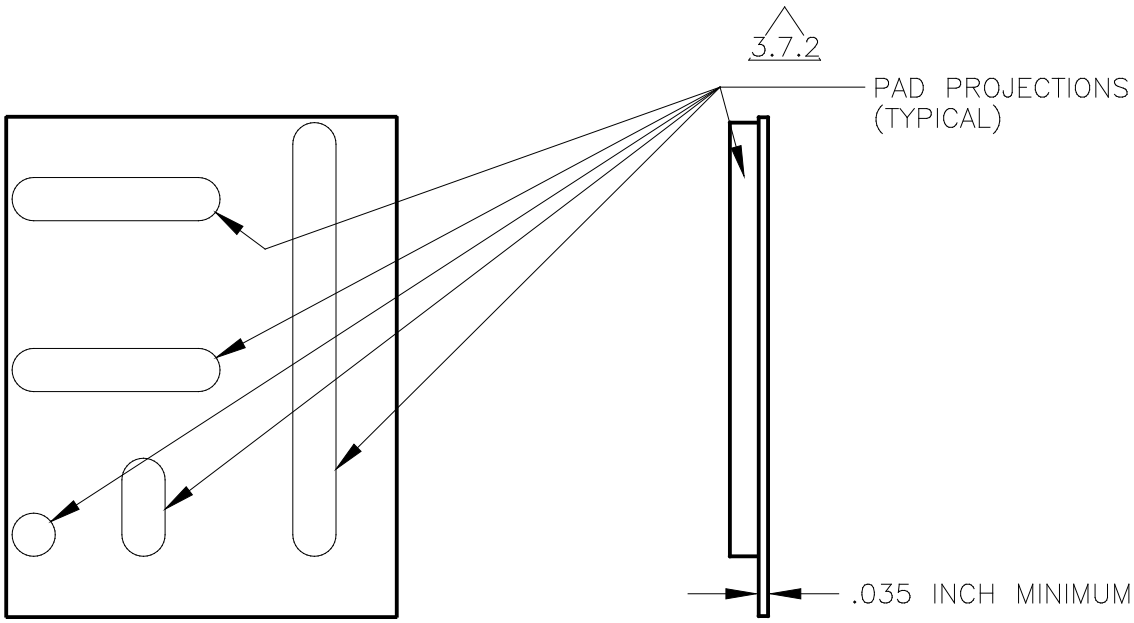


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 2. Cover for PIN M81757/11-3.

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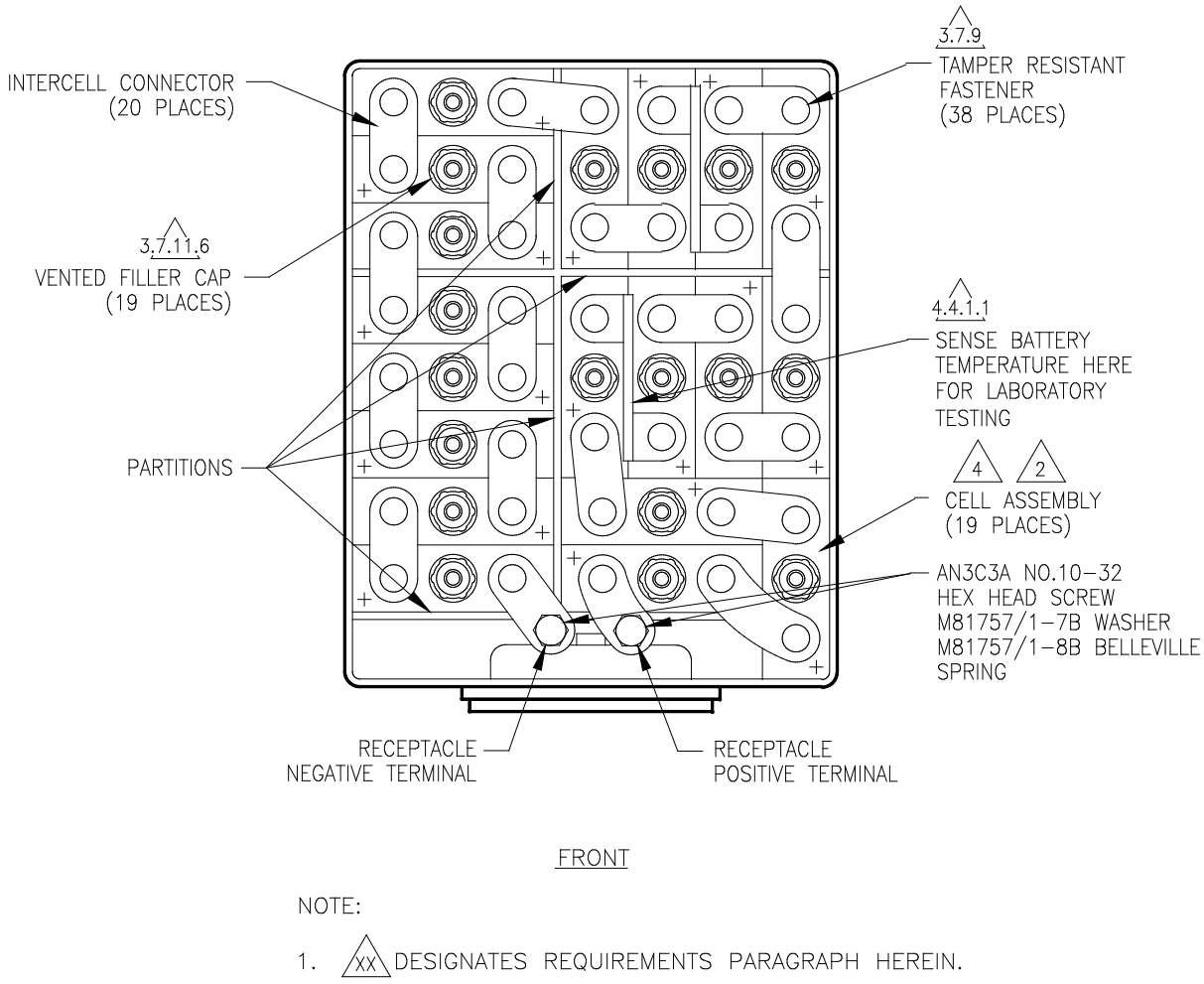


NOTES:

1. THE PAD SHALL CONFORM TO 3.4, 3.7.2 AND 3.7.3.
2. WHEN COVER IS SECURED, THE PROJECTIONS SHALL CONTACT EACH INTERCELL CONNECTOR AND SHALL NOT HINDER THE PROPER RELEASE OF GASES.

FIGURE 3. Cover gasket and cell hold-down pad for M81757/11-3.

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NOTE:

1. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 4. Layout of interior components for PIN M81757/11-3.

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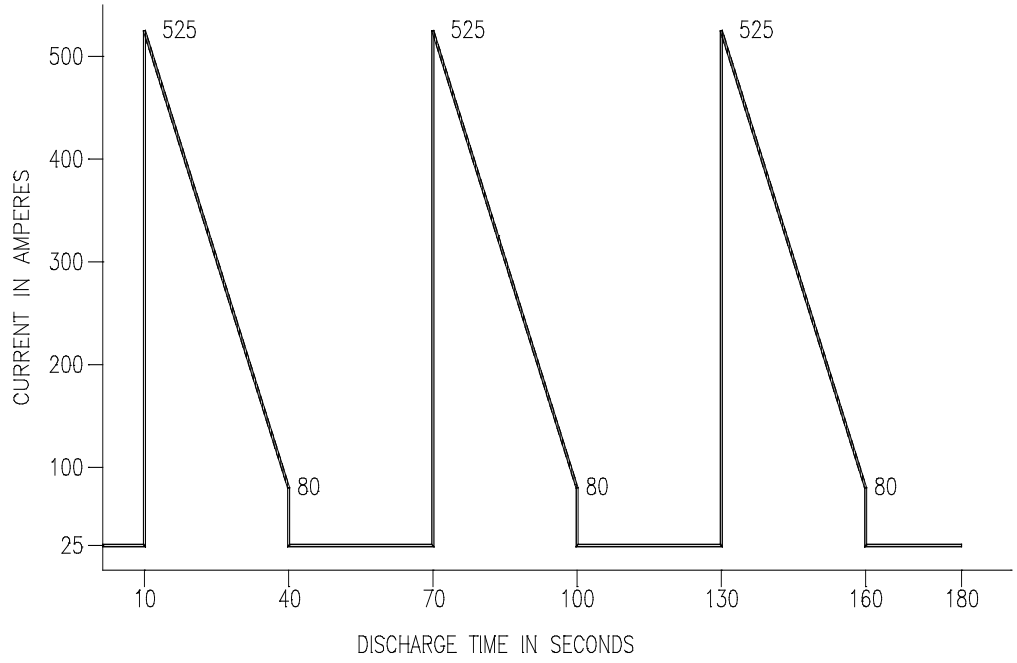


FIGURE 5. Slope discharge profile.

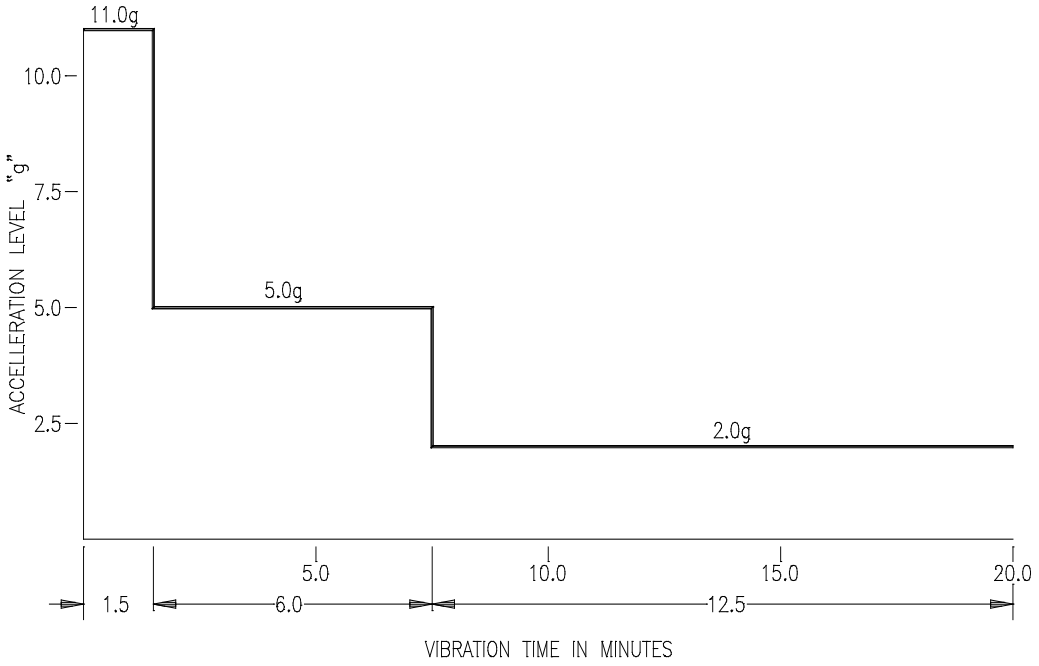


FIGURE 6. Low frequency (17 Hz) vibration for 20 minutes.

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NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, LIMITED-REPAIR, LOW-FREQUENCY VIBRATION
MILITARY PIN:	M81757/11-3
NATIONAL STOCK NUMBER:	6140-**-***-****
MAXIMUM WEIGHT:	55 LB
NOMINAL VOLTAGE:	24 VOLTS
MINIMUM CAPACITY:	20 AH (20 AMPS/1 HR/80°F/18 V)
CONTRACT AND DELIVERY ORDER NUMBER:	***
MANUFACTURED BY:	***
MANUFACTURER'S CAGE:	***
MANUFACTURER'S SERIAL NUMBER:	***
DATE/LOT CODE:	***
DATE FIRST PLACED IN SERVICE:	

FIGURE 7. Marking for M81757/11-3 batteries.

	MINIMUM LETTERING HEIGHT
CAUTION The cells in this battery are not to be removed. This battery shall be serviced only in an authorized nickel-cadmium battery shop. ⇩ RECEPTACLE ⇩	0.50 ± 0.03 inch lettering height. 0.375 ± 0.03 inch lettering height. 0.375 ± 0.03 inch lettering height 0.50 ± 0.03 inch lettering height

FIGURE 8. Marking for battery caution and receptacle.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery shall be M81757/11-3.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8 as modified herein and as shown on figures 1, 2, and 3 herein.

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3. Capacity. The minimum capacity of the battery shall be not less than 20 Ah. See 3.10.14 as modified herein.

4. Interface requirements. The cells and other interior components of the battery shall be arranged and positioned as shown on figure 4 herein. The cell terminals shall be vertically and horizontally positioned so that each cell terminal shall mate with its matching electrical connection of the cell scanner fixture and cell equalization fixture. The components and fasteners specified on figure 4 are required due to interfaces with the MS3509-31 receptacle, the cell scanner fixture, and the cell equalization fixture.

5. Repairability. The M81757/11-3 is a limited-repair battery. The maintenance/repair procedures and practices for the M81757/11-3 battery shall not include removing and replacing the battery's cells.

6. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

6.1 Modify the following paragraphs:

3.7.2 Battery container and cover. In the fourth sentence, delete “the applicable specification sheet” and insert “figures 1 and 2 of the specification sheet.” Delete the fifth sentence and insert: “The cover shall be fitted with a hold-down pad that functions as the cover’s gasket and shall have integral projections that hold down each battery cell. The cover gasket/cell hold-down pad shall conform to figure 3 of the specification sheet. The cover pad/gasket shall be sized and fitted to match the rim of the container body for a gastight fit and effective fluid seal between the cover and the container body. The location of the projections shall be determined by the location of each intercell connector. The projections shall contact each intercell connector and apply a force on each intercell connector to restrain vertical movement of the cells during vibration in accordance with 4.5.16 and 4.5.32. An insulating liner 0.062 ± 0.007 inch thick shall be contained between the inside, top surface of the container cover and the cover pad/gasket. The liner shall provide complete coverage of the inside, top surface of the container cover and shall conform to the requirements of 3.10.3 or 3.10.4, as applicable (see 6.10.2).”

3.7.5 Handles. Not applicable.

3.7.8 Receptacles. Following the last sentence, insert: "The MS3509-31 receptacle shall be vertically positioned so that the contact surface of each of the two terminal blocks is in the same plane as the contact surface of their respective interconnecting cell terminal (see figures 1 and 4)."

3.7.11.6 Vented filler cap. Following the first sentence, insert: "Each cell shall contain a vented filler cap conforming to PIN M81757/2-2 for the M81757/11-3 battery."

3.8.4.1 Limited-repair batteries. In the first sentence, change “figure 1” to “figure 7 of the specification sheet.”

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3.8.4.3 Battery caution and receptacle marking. In the second and third sentences, change "figure 3" to "figure 8 of the specification sheet."

3.10.8 Dimensions and weights. Delete the entire text and substitute: "The dimensions and configuration of the battery shall be as shown on figures 1, 2, 3, and 4 of the specification sheet. The weight of the battery shall be 53.5 ±1.5 pounds with the electrolyte level adjusted in accordance with 4.4.4. All dimensions shall be in inches and prior to coating except overall width (10.148 max), overall length (10.662 max), overall height (8.812 max), container base width (7.838 +0.062, -0.15), and container base length (9.938 ± 0.062). The thickness of the spacers shall be not greater than 0.015 inch. The thickness of the battery's container's sides and bottom shall be not less than 16 gauge; the front and back, not less than 20 gauge; and the container cover, not less than 18 gauge."

3.10.10 Handle strength. Not applicable.

3.10.14 Capacity. Delete the text of this paragraph and all subparagraphs and insert: "The minimum capacity of the battery shall be not less than 20 Ah. To meet the indicated capacity requirement, the battery shall produce a potential of not less than 18.0 volts at not less than 60 minutes into the C-rate discharge after being tested in accordance with 4.5.13 at room temperature (see 4.4.1)."

3.10.22 Constant voltage discharge. Not applicable.

3.10.25 Physical integrity at high temperature. In the first sentence, delete "and handles."

6.2 Add the following paragraphs:

"3.10.31 Slope discharge. The battery shall perform as follows after being tested in accordance with 4.5.31.

"a. The battery shall produce a potential of not less than 12.0 volts after being discharged in accordance with figure 5 of the specification sheet for two consecutive discharges.

"b. The battery shall also meet the requirements of 3.9 during and after testing and discharge.

"3.10.32 Low frequency vibration. The battery shall perform as follows after being tested in accordance with 4.5.32.

"a. During the charging of the battery in the test of 4.5.32, the increase in the charge current shall be not greater than 5 amperes from the lowest value reached.

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"b. After being tested in accordance with 4.5.32, the battery shall produce a potential of not less than 12.0 volts when discharged in accordance with figure 6 of the specification sheet during each of the 17 vibration cycles.

"c. The battery shall also meet the requirements of 3.9 during and after testing and discharge."

"4.4.1.1 Temperature measurement location. When the battery temperature is to be measured during laboratory testing, the temperature shall be measured at the location between the two cells shown on figure 4 of the specification sheet. The temperature shall be measured with a temperature sensor (thermocouple or thermistor) inserted between the two cells and positioned in the approximate center of the outer sidewall of each cell equidistant from top to bottom and side to side."

"4.4.10 Mounting for vibration testing. The battery shall be clamped in the cradle by two flat bars, with one of the bars bearing on the cover adjacent to each of the left and right edges of the cover, and extending inward from the edges 0.844 ± 0.100 inch from the sides, measured in the lower 2/10 of the height, and extending from the front edge of the cover to the back edge of the cover. The hold-down bars shall be clamped to the cradle by four vertical 5/16-24 threaded rods with nuts tightened to 45 ± 3 inch-pounds, bearing on the hold-down bars through cross-beams. The rods shall be arranged symmetrically about the battery, with two rods on each side of the battery. Two of the rods shall be 2.500 ± 0.100 inches from the front of the battery, and two of the rods shall be 2.500 ± 0.100 inches from the back of the battery, measured in the lower 2/10 of the height of the battery."

6.3 Modify the following paragraphs:

4.5.9 Handle strength test. Not applicable.

4.5.22 Constant voltage discharge. Not applicable.

4.5.23 Cycling test. Delete the text of 4.5.23.b(1) and insert: "(1) Discharge the battery for 20 seconds through a 30-milliohm resistance."

4.5.25 Physical integrity at high temperature. In step c, delete the second sentence.

6.4 Add the following paragraphs:

"4.5.31 Slope discharge test. This test shall consist of the following steps.

"a. Place the fully charged battery in a 38° ± 2°C (100° ± 5°F) environment and allow its temperature to stabilize.

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"b. While the battery remains within the temperature environment of 4.5.31.a, discharge the battery in accordance with figure 5 of this specification sheet.

"c. Place the battery in a room temperature environment and constant potential charge the battery at 27.74 ± 0.10 volt for 5.0 ± 0.1 hours.

"d. Repeat 4.5.31.a and 4.5.31.b.

"e. Continuously measure the terminal voltage and temperature of the battery.

"f. Examine the battery for the requirements of 3.10.31."

"4.5.32 Low frequency vibration test. The battery shall not receive any maintenance throughout this test. If the test battery's terminals are an open circuit for not less than 24 hours after starting and before completing this test, the battery shall be constant potential charged in accordance with 4.4.2 before continuing the test. This test shall consist of the following steps.

"a. Remove the cover from the battery.

"b. Constant potential charge the battery in accordance with 4.4.2.

"c. Vibration cycle: The following steps comprise one vibration cycle. Steps 4.5.32.c(3) and 4.5.32.c(4) will amount to a total of 5 hours of vibration.

"(1) Place the battery in a 38° ± 2°C (100° ± 5°F) environment and allow its temperature to stabilize. Perform the discharge of figure 5 of the specification sheet.

"(2) Remove the battery from the 38° ± 2°C (100° ± 5°F) environment, secure the cover onto the battery, and proceed immediately with 4.5.33.c(3).

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"(3) Attach the battery (oriented in its normal upright position) to the vibration equipment (see 4.4.10). Ventilate the battery through the vent tubes at 1 ± 10 percent cubic feet per minute during vibration. Vibrate the battery in accordance with figure 6 of the specification sheet. Constant potential charge the battery at 27.74 ± 0.10 volts while the battery is subjected to vibration.

"(4) Repeat 4.5.33.c(3) fourteen times.

"d. Repeat 4.5.32.c sixteen times. The seventeen vibration cycles will amount to a total of 85 hours of vibration. Step 4.5.32.c(3) shall be repeated not more than two times within a 24-hour period.

"e. Continuously measure the battery's terminal voltage and temperature and the charging current.

"f. Examine the battery for the requirements of 3.9 and 3.10.32."

6.5 Modify the following paragraphs and tables:

6.1 Intended use. Add the following after the last sentence: "The M81757/11-3 limited-repair battery is designed so that its cells and all components associated with installing the cells cannot be removed and replaced. The battery should be requisitioned and used only by those DoD Departments and Agencies whose aircraft battery maintenance/repair policies, procedures, and practices disallow the battery's cells to be removed and replaced."

6.10.3 Handles. Not applicable.

6.15 PIN supersession data. Delete text and insert: "Battery PIN M81757/11-3 supersedes the batteries and cells shown in tables IX and X."

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TABLE III. Qualification inspection of batteries.

Delete test number 5 (Handle Strength Test) and test number 18 (Constant Voltage Discharge). Add the following test:

TEST NO.	EXAMINATIONS AND TESTS	SAMPLE NUMBER					REQUIREMENT PARAGRAPH	TEST PARAGRAPH
		1	2	3	4	5		
26a	LOW FREQUENCY VIBRATION TEST	X					3.9, 3.10.32	4.5.32

TABLE VIII. Group C inspection.

Delete tests II-6 (Humidity and charge retention) and III-6 (Constant Voltage Discharge). Add the following test:

TEST NO.	EXAMINATIONS AND TESTS	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
	SUBGROUP I:		
I-6a	SLOPE DISCHARGE TEST	3.9, 3.10.31	4.5.31

TABLE IX. Battery and cell supersession data.

Delete the title and contents and substitute the following table:

Superseding PIN (battery)	Superseded PINs (batteries/cells)
M81757/11-3	MS90377-1 and M81757/11-1 (batteries) MS90321-79W (cell) 1/
None (cancelled)	M81757/11-2 and M81757/11-4 (batteries; cancelled)

1/ Substitute the M81757/11-3 battery for each quantity of 19 or fewer cells ordered.

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6.6 Add the following table:

TABLE X. Superseded (replaced) batteries and cells.

Superseded PIN (battery/cell)	General specification or procurement specification of superseded PIN	Superseding PIN (battery)
MS90321-79W (cell) 1/	MIL-PRF-8565	M81757/11-3
MS90377-1 (battery)	MIL-PRF-8565	M81757/11-3
M81757/11-1 (battery)	MIL-PRF-81757	M81757/11-3
M81757/11-2 (battery)	MIL-PRF-81757	None (cancelled)
M81757/11-4 (battery)	MIL-PRF-81757	None (cancelled)

1/ Substitute the M81757/11-3 battery for each quantity of 19 or fewer cells ordered.

7. Referenced documents. In addition to MIL-PRF-81757, this specification sheet references MIL-PRF-8565.

8. Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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DLA - CC

Preparing activity:
Navy - AS

Agent:
Navy - SH
Project 6140-2021-007

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